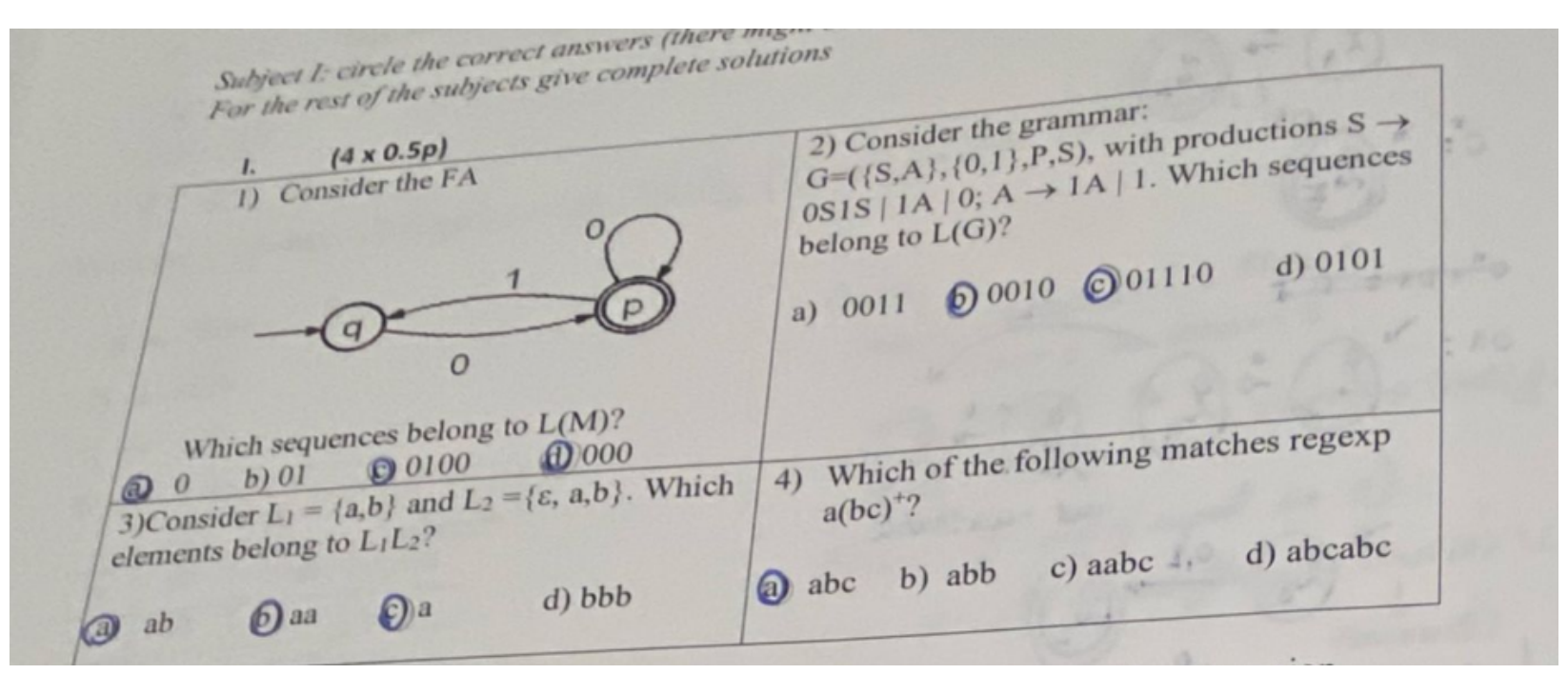
DTicket 4

24.01.2025 14:38



Subject I: circle the correct answers (there mig
For the rest of the subjects give complete solutions

I. ***(4 x 0.5p)***

1) Consider the FA

Which sequences belong to L(M)?

[✓ a)] 0 b) 01 [✓ c)] 0100 [✓ d)] 000

2) Consider the grammar: G=({S,A},{0,1},P,S), with productions S → 0S1S | 1A | 0; A → 1A | 1. Which sequences belong to L(G)?

a) 0011 [✓ b)] 0010 [✓ c)] 01110 d) 0101

3)Consider $L_1 = \{a,b\}$ and $L_2 = \{\varepsilon, a,b\}$. Which elements belong to L_1L_2?

[✓ a)] ab [✓ b)] aa [✓ c)] a d) bbb

4) Which of the following matches regexp $a(bc)^+$?

[✓ a)] abc b) abb c) aabc d) abcabc

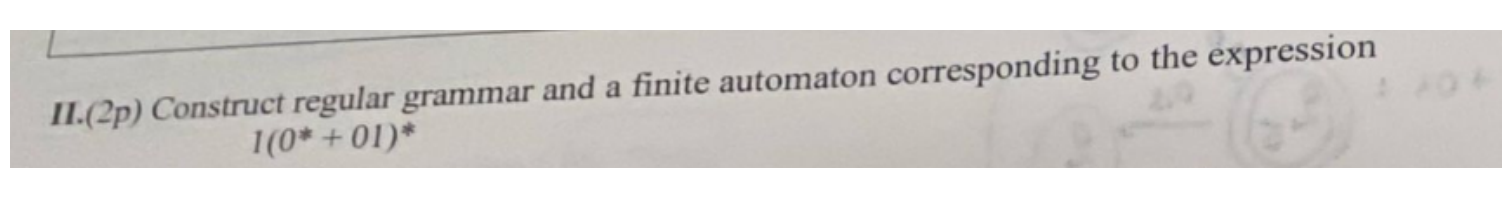
1) a, c, d

2) S → 0S1S | 1A | 0
A → 1A | 1

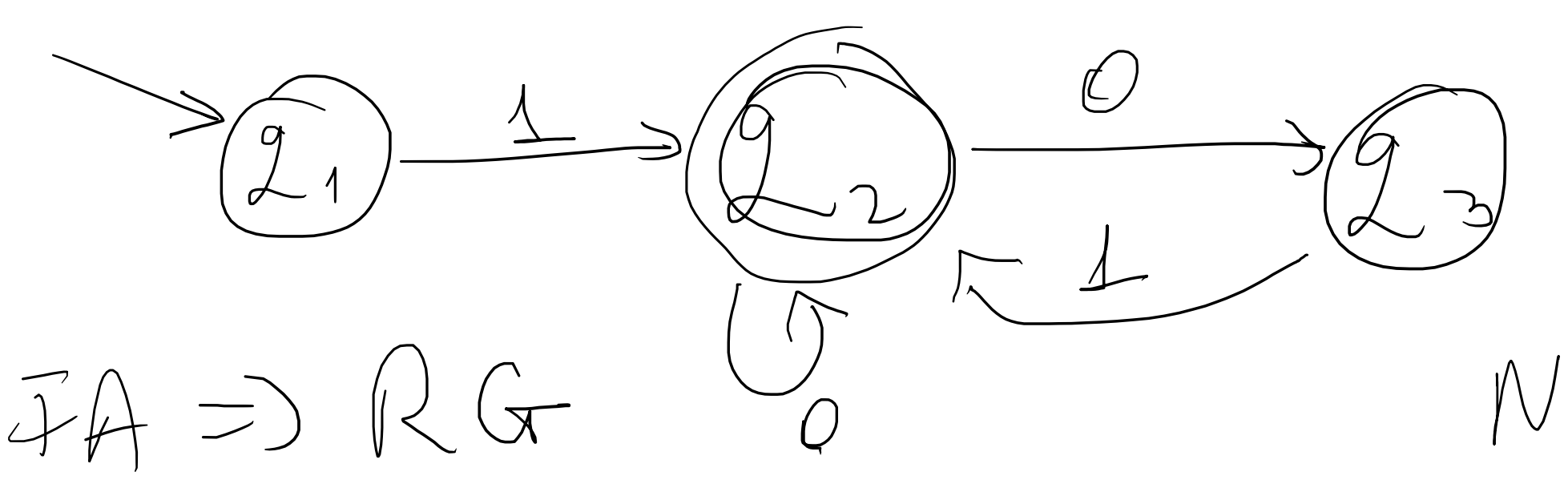
b, c

3) a, b, c

4) a



II.(2p) *Construct regular grammar and a finite automaton corresponding to the expression*
$1(0^* + 01)^*$



FA ⇒ RG

$N = \{q_1, q_2, q_3\}$, $\Sigma = \{0, 1\}$

$\delta(q_1, 1) = q_2$

$\delta(q_2, 0) = q_2$

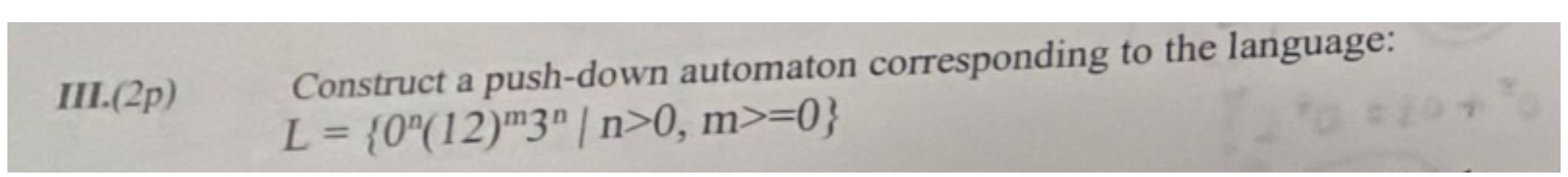
$\delta(q_2, 0) = q_3$

$\delta(q_3, 1) = q_2$

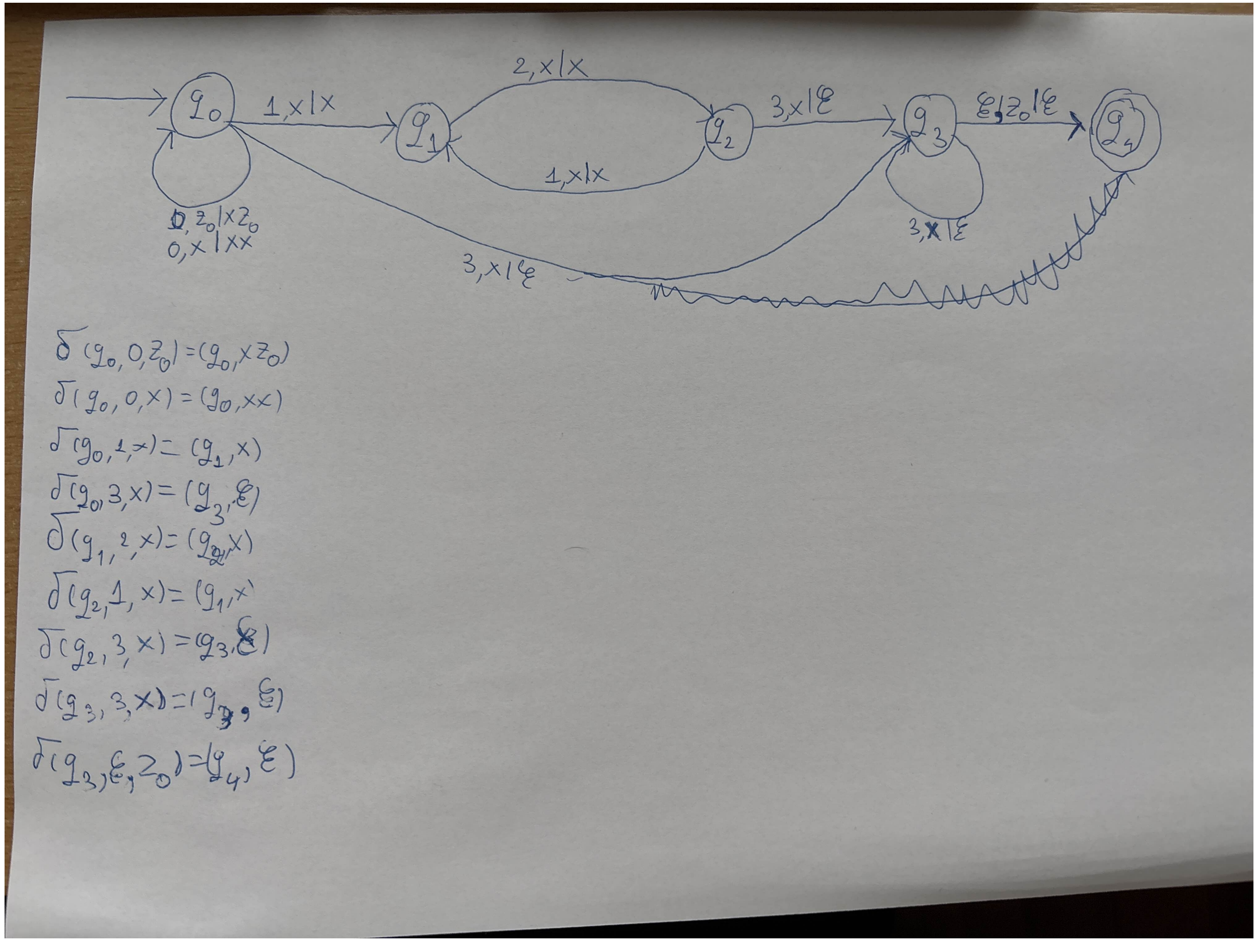
$q_1 \to 1q_2 \mid 1$

$q_2 \to 0q_2 \mid 0q_3 \mid 0$

$q_3 \to 1q_2 \mid 1$



III.(2p) Construct a push-down automaton corresponding to the language:
$L = \{0^n(12)^m3^n \mid n>0, m>=0\}$



$\delta(q_0, 0, Z_0) = (q_0, xZ_0)$

$\delta(q_0, 0, x) = (q_0, xx)$

$\delta(q_0, 1, x) = (q_1, x)$

$\delta(q_0, 3, x) = (q_3, \varepsilon)$

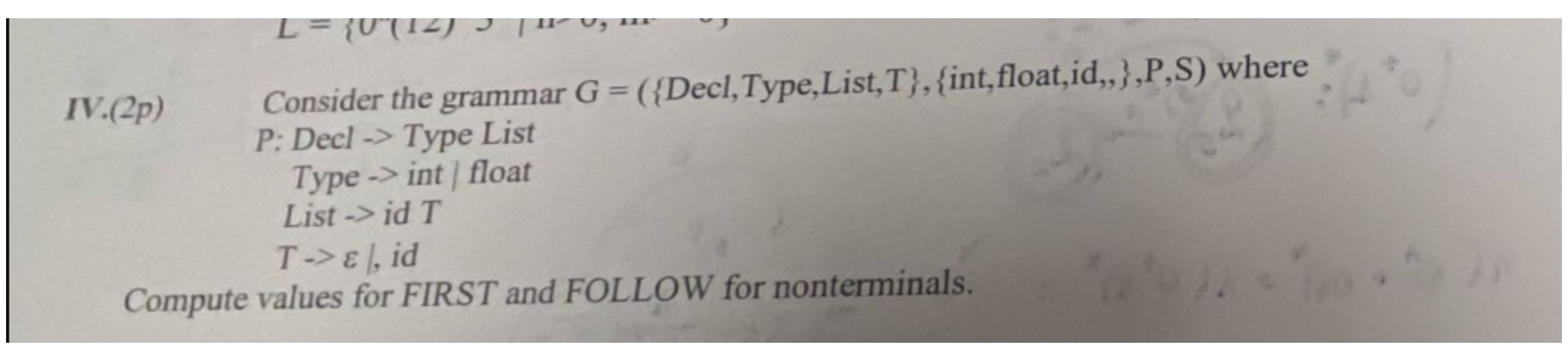
$\delta(q_1, 2, x) = (q_2, x)$

$\delta(q_2, 1, x) = (q_1, x)$

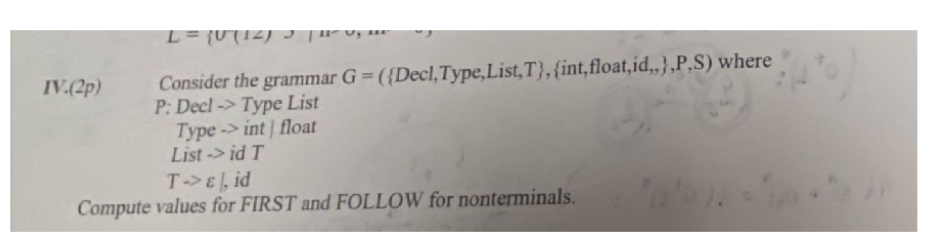
$\delta(q_2, 3, x) = (q_3, \varepsilon)$

$\delta(q_3, 3, x) = (q_3, \varepsilon)$

$\delta(q_3, \varepsilon, Z_0) = (q_4, \varepsilon)$



IV.(2p) *Consider the grammar G = ({Decl,Type,List,T},{int,float,id,,},P,S) where*
P: Decl -> Type List
Type -> int | float
List -> id T
T -> ε |, id
Compute values for FIRST and FOLLOW for nonterminals.



IV.(2p) Consider the grammar G = ({Decl,Type,List,T},{int,float,id,,},P,S) where
P: Decl -> Type List
Type -> int | float
List -> id T
T -> ε |, id
Compute values for FIRST and FOLLOW for nonterminals.

A → B.. | C.. | D..

1. Decl → Type List
2/3. Type → int | float
4. List → id T
5/6. T → ε | , id

First:

Decl	∅	int, float
Type	int, float	int, float
List	id	id
T	ε, ,	ε, ,

¬F(S) ∋ ε

- iei productile
- pentru fiecare productie iei pe rand non-terminalii din ea si pentru fiecare non-terminal A, daca apare in productie ceva dupa el (AB) -> la Follow(A) adaugi First(B)
- daca nu mai apare nimic (e ultimul din productie), la Follow(A) adaugi Follow(S), unde S e non-terminalul care a generat productia

Follow:

Decl	ε
Type	id
List	ε
T	ε

Decl → Type List
Type → int | float
List → id T
T → ε | , id

$\boxed{\text{int a,b}}$

	int	float	,	id
Decl	Type List	Type List	-	—
Type	int	float	—	—
List	—	—	—	id T
T			, id	—

First (Prod) ∋ t,
← t ∈ First (NT)
$^{+}$(NT, t) = Prod

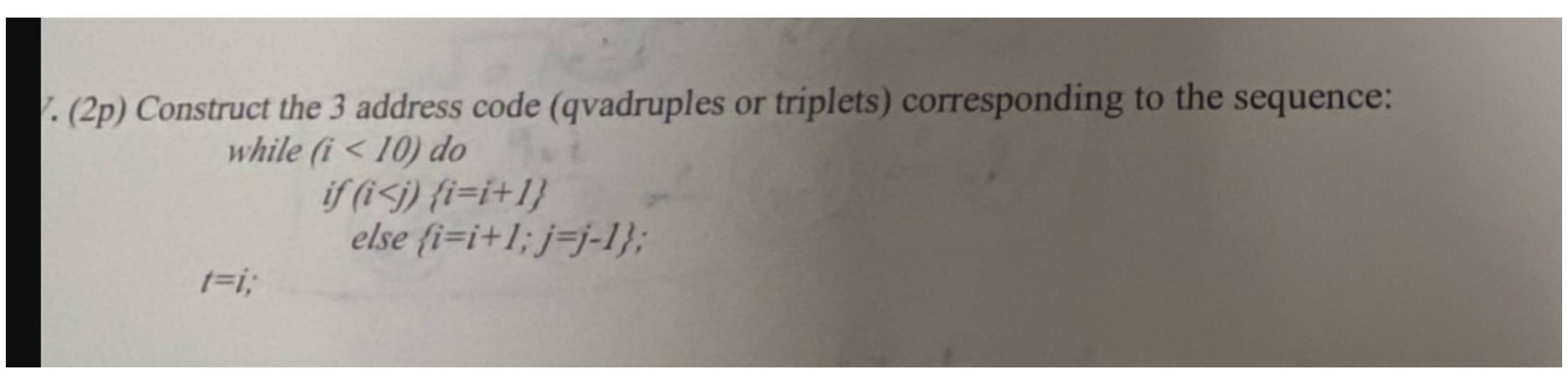
$(\text{Decl}\,\$,\ \text{int a,b}\,\$,\ \varepsilon) \vdash (\text{Type List}\,\$,\ \text{int a,b}\,\$,\ 1) \vdash$

$\vdash (\text{int List}\,\$,\ \text{int a,b}\,\$,\ 12) \overset{pop}{\vdash} (\text{List}\,\$,\ \text{a,b}\,\$,\ 12) \vdash$

$\vdash (\text{id T}\,\$,\ \text{a,b}\,\$,\ 124) \overset{pop}{\vdash} (\text{T}\,\$,\ \text{,b}\,\$,\ 124) \vdash (\text{,id}\,\$,\ \text{,b}\,\$,\ 1246)$

$\overset{pop}{\vdash} (\text{id}\,\$,\ \text{b}\,\$,\ 1246) \overset{pop}{\vdash} (\$,\ \$,\ 1246)$

1. $M(A,a) = (\alpha, i), \forall a \in FIRST(\alpha), a \neq \epsilon, A \to \alpha$ production in P with number i
   $M(A,b) = (\alpha, i),$ if $\epsilon \in FIRST(\alpha), \forall b \in FOLLOW(A), A \to \alpha$ production in P with number i
2. $M(a,a) = pop, \forall a \in \Sigma;$
3. $M(\$,\$) = acc;$
4. M(x,a)=err (error) otherwise i.



(2p) Construct the 3 address code (qvadruples or triplets) corresponding to the sequence:

```
while (i < 10) do
    if (i<j) {i=i+1}
     else {i=i+1; j=j-1};
t=i;
```

label	op	arg1	arg2	res
1	≥	i	10	t1
2	goto	t1		(1)
3	<	i	j	t2
4	goto	t2		(10)
5	+	i	1	t3
6	=	t3		i
7	-	j	1	t4
8	=	t4		j
9	goto			(1)
10	+	i	1	t5
11	=	t5		i
12	goto			(1)
13	=	i		t